

bn

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1 basics

$$\begin{aligned} p(A, D) &= \sum_{B, C, E, F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\ &= \sum_{B, C} p(A)p(B|A)p(C|B, D)p(D) = \sum_B p(A)p(B|A)p(D) \\ &= p(A)p(D), \end{aligned}$$

so $A \perp D$.

$$\begin{aligned} p(B, D) &= \sum_{A, C, E, F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\ &= \sum_{A, C} p(A)p(B|A)p(C|B, D)p(D) \\ &= \sum_A p(A)p(B|A)p(D) = p(B)p(D). \end{aligned}$$

$$\begin{aligned} p(A, B) &= \sum_{C, D, E, F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\ &= \sum_{C, D} p(A)p(B|A)p(C|B, D)p(D) = \sum_D p(A)p(B|A)p(D) \\ &= p(A)p(B|A) = p(A, B). \end{aligned}$$

$$\begin{aligned} p(B, E|A) &= \frac{p(A, B, E)}{p(A)} = \frac{1}{p(A)} \sum_{C, D, F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\ &= \frac{1}{p(A)} \sum_D p(A)p(B|A)p(D)p(E|A) = \frac{1}{p(A)} p(A)p(B|A)p(E|A) \\ &= p(B|A)p(E|A), \end{aligned}$$

so $B \perp E|A$.

$$\begin{aligned}
p(C, E|A) &= \frac{1}{p(A)} \sum_{B,D,F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\
&= \frac{1}{p(A)} \sum_{B,D} p(A)p(B|A)p(C|B, D)p(D)p(E|A) \\
&= p(E|A) \sum_{B,D} p(B|A)p(C|B, D)p(D) \\
&= \frac{p(E|A)}{p(A)} \sum_{B,D} p(A, B, D)p(C|B, D)
\end{aligned}$$

$$\begin{aligned}
p(A, C|B) &= \frac{1}{p(B)} \sum_{D,E,F} p(A)p(B|A)p(C|B, D)p(D)p(E|A)p(F|B, C) \\
&= \frac{1}{p(B)} \sum_D p(A)p(B|A)p(C|B, D)p(D) \\
&= \frac{1}{p(B)} \sum_D p(A, B) \frac{p(B, C, D)}{p(B, D)} p(D) \\
&= \frac{1}{p(B)} \sum_D p(A, B) \frac{p(B, C, D)}{p(B)} = \frac{1}{p(B)^2} p(A, B)p(B, C) = p(A|B)p(C|B),
\end{aligned}$$

so $A \perp C|B$.